Hospital Clinical Quality & Emergency Department Performance Analysis

Project Overview

This project analyzes publicly available hospital-level performance data from the Centers for Medicare & Medicaid Services (CMS).

Business Objectives

- Assess the completeness and consistency of hospital performance data.
- Explore how records are distributed across states, conditions, and quality measures.
- Analyze a selected emergency department measure(OP_18a).
- Compare reported performance across states.
- Identify unusual observations that may need further investigation.
- Produce clean datasets and findings to support healthcare reporting.

Tools

- Python
- pandas
- Matplotlib
- Seaborn
- Jupyter Notebook

Scope and Limitations

This analysis uses hospital reporting data. Statistical patterns and outliers are descriptive findings. They do not prove that a hospital provides better or worse care.

Import, load data, and initial exploration

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv(
    "Timely_and_Effective_Care-Hospital.csv",
    dtype={"Facility ID": "string", "ZIP Code": "string"},
    low_memory=False
)
```

```
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])

df.head()

df.info()

print(df.describe(include="all").T)

missing_summary = pd.DataFrame({
    "Missing Count": df.isna().sum(),
    "Missing Percent": (df.isna().mean() * 100).round(2)
}).sort_values("Missing Percent", ascending=False)

print(missing_summary)

print("Unique hospitals:", df["Facility ID"].nunique())
print("Unique states:", df["State"].nunique())
print("Unique conditions:", df["Condition"].nunique())
print("Unique measures:", df["Measure ID"].nunique())

# print("\nScore values:")
# print(sorted(df["Score"].dropna().astype(str).unique()))

# print("\nSample values:")
# print(sorted(df["Sample"].dropna().astype(str).unique()))
```

```
Rows: 138084
Columns: 16
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 138084 entries, 0 to 138083
Data columns (total 16 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   Facility ID       138084 non-null  string
 1   Facility Name     138084 non-null  object
 2   Address           138084 non-null  object
 3   City/Town         138084 non-null  object
 4   State             138084 non-null  object
 5   ZIP Code          138084 non-null  string
 6   County/Parish     138084 non-null  object
 7   Telephone Number  138084 non-null  object
 8   Condition         138084 non-null  object
 9   Measure ID        138084 non-null  object
 10  Measure Name      138084 non-null  object
 11  Score             138084 non-null  object
 12  Sample            133426 non-null  object
 13  Footnote          91835 non-null   object
 14  Start Date        138084 non-null  object
 15  End Date          138084 non-null  object
dtypes: object(14), string(2)
memory usage: 16.9+ MB
```

	count	unique	top	freq
Facility ID	138084	4658	670322	30
Facility Name	138084	4538	MEMORIAL HOSPITAL	353
Address	138084	4624	100 HOSPITAL DRIVE	180
City/Town	138084	2910	CHICAGO	849
State	138084	56	TX	11760
ZIP Code	138084	4308	77030	173
County/Parish	138084	1549	LOS ANGELES	2206
Telephone Number	138084	4625	(469) 341-7800	135
Condition	138084	6	Electronic Clinical Quality Measure	68214
Measure ID	138084	30	EDV	4658
Measure Name	138084	30	Emergency department volume	4658
Score	138084	685	Not Available	80338
Sample	133426	8631	Not Available	78819
Footnote	91835	17	5	63815
Start Date	138084	2	01/01/2024	86846
End Date	138084	3	12/31/2024	86846

	Missing Count	Missing Percent
Footnote	46249	33.49
Sample	4658	3.37
Facility ID	0	0.00
Facility Name	0	0.00
State	0	0.00
ZIP Code	0	0.00
Address	0	0.00
City/Town	0	0.00
Telephone Number	0	0.00
County/Parish	0	0.00
Condition	0	0.00
Measure ID	0	0.00
Score	0	0.00

```
Measure Name                0             0.00
Start Date                  0             0.00
End Date                    0             0.00
Unique hospitals: 4658
Unique states: 56
Unique conditions: 6
Unique measures: 30
```

Data cleaning and standardization

In [2]:
```python
df.columns = df.columns.str.strip()

text_columns = [
    "Facility ID", "Facility Name", "Address", "City/Town",
    "State", "ZIP Code", "County/Parish", "Telephone Number",
    "Condition", "Measure ID", "Measure Name", "Score",
    "Sample", "Footnote"
]

for col in text_columns:
    df[col] = df[col].astype("string").str.strip()

# Convert empty strings into actual missing values
df[text_columns] = df[text_columns].replace(
    r"^\s*$", pd.NA, regex=True
)

# Standardize state codes
df["State"] = df["State"].str.upper()

# Convert scores to numeric where possible
df["Score"] = pd.to_numeric(df["Score"], errors="coerce")

# Convert sample size to numeric where possible
df["Sample"] = pd.to_numeric(df["Sample"], errors="coerce")

# Parse reporting dates
for col in ["Start Date", "End Date"]:
    df[col] = pd.to_datetime(
        df[col],
        format="%m/%d/%Y",
        errors="coerce"
    )

df[["Facility ID", "State", "Measure ID",
    "Score", "Start Date", "End Date"]].head()
```

Out[2]:

	Facility ID	State	Measure ID	Score	Start Date	End Date
0	010001	AL	EDV	<NA>	2024-01-01	2024-12-31
1	010001	AL	GMCS	<NA>	2024-01-01	2024-12-31
2	010001	AL	GMCS_Malnutrition_Diagnosis_Documented	<NA>	2024-01-01	2024-12-31
3	010001	AL	GMCS_Malnutrition_Screening	<NA>	2024-01-01	2024-12-31
4	010001	AL	GMCS_Nutrition_Assessment	<NA>	2024-01-01	2024-12-31

Duplicate detection and data quality checks

In [3]:
```
exact_duplicates = df.duplicated().sum()

print("Exact duplicate rows:", exact_duplicates)

df = df.drop_duplicates().copy()

print("Shape after removing exact duplicates:", df.shape)

record_key = [
    "Facility ID",
    "Measure ID",
    "Start Date",
    "End Date"
]

duplicate_keys = df.duplicated(
    subset=record_key,
    keep=False
)

print("Rows sharing a hospital-measure-period key:",
      duplicate_keys.sum())

print(
    df.loc[duplicate_keys]
      .sort_values(record_key)
      .head(20)
)

df["Score Original"] = df["Score"]

df["Score Numeric"] = pd.to_numeric(
    df["Score"],
    errors="coerce"
```

```
)

print(
    df["Score"]
    .value_counts(dropna=False)
    .head(30)
)

# Values that could not be converted to numbers
nonnumeric_scores = df.loc[
    df["Score"].notna() &
    df["Score Numeric"].isna(),
    ["Measure ID", "Measure Name", "Score", "Footnote"]
]

print("Non-numeric score rows:", len(nonnumeric_scores))

print(nonnumeric_scores.head(20))

print(
    "Rows with missing or nonnumeric scores:",
    df["Score Numeric"].isna().sum()
)

print(
    "Percentage without numeric scores:",
    round(df["Score Numeric"].isna().mean() * 100, 2),
    "%"
)

print("Missing Sample:", df["Sample"].isna().sum())
print("Missing Footnote:", df["Footnote"].isna().sum())

print("\nFootnote codes:")
print(df["Footnote"].value_counts(dropna=False).head(20))

# Missing identifiers
print("Missing facility IDs:", df["Facility ID"].isna().sum())
print("Missing measure IDs:", df["Measure ID"].isna().sum())

# Invalid reporting periods
invalid_dates = df[
    df["Start Date"].isna() |
    df["End Date"].isna() |
    (df["Start Date"] > df["End Date"])
]

print("Invalid reporting periods:", len(invalid_dates))

# Inspect numeric score ranges by measure
score_ranges = (
    df.dropna(subset=["Score Numeric"])
      .groupby(["Measure ID", "Measure Name"])["Score Numeric"]
      .agg(["count", "min", "median", "max"])
      .sort_values("count", ascending=False)
)
```

```
print(score_ranges.head(10))
```

```
Exact duplicate rows: 0
Shape after removing exact duplicates: (138084, 16)
Rows sharing a hospital-measure-period key: 0
Empty DataFrame
Columns: [Facility ID, Facility Name, Address, City/Town, State, ZIP Code, County/Pa
rish, Telephone Number, Condition, Measure ID, Measure Name, Score, Sample, Footnot
e, Start Date, End Date]
Index: []
Score
<NA>    84164
100      2120
0        2046
1        1998
98       1453
97       1413
99       1281
96       1268
95       1172
2        1126
94       1056
92        996
93        993
91        820
88        799
90        764
89        687
86        626
85        610
87        602
83        598
84        562
3         543
81        535
82        533
80        530
79        517
76        474
75        447
78        443
Name: count, dtype: Int64
Non-numeric score rows: 0
Empty DataFrame
Columns: [Measure ID, Measure Name, Score, Footnote]
Index: []
Rows with missing or nonnumeric scores: 84164
Percentage without numeric scores: 60.95 %
Missing Sample: 83477
Missing Footnote: 46249

Footnote codes:
Footnote
5          63815
<NA>       46249
2          10583
1           9033
19          3916
```

```
1, 3        1578
4            627
3            541
1, 2, 3      530
1, 2         292
7            281
29           224
3, 7         171
2, 3         157
1, 29         52
2, 3, 7       20
2, 7          12
12             3
Name: count, dtype: Int64
Missing facility IDs: 0
Missing measure IDs: 0
Invalid reporting periods: 0
```

Measure ID	Measure Name	count
IMM_3	Healthcare workers given influenza vaccination	4260
OP_18b	Average (median) time patients spent in the eme...	4081
OP_18a	Average (median) time all patients spent in the...	4052
OP_22	Left before being seen	3821
SAFE_USE_OF_OPIOIDS	Safe Use of Opioids - Concurrent Prescribing	3740
SEP_1	Appropriate care for severe sepsis and septic s...	3115
SEV_SEP_3HR	Severe Sepsis 3-Hour Bundle	3094
OP_18c	Average (median) time psychiatric/mental health...	3023
SEV_SEP_6HR	Severe Sepsis 6-Hour Bundle	2957
OP_29	Endoscopy/polyp surveillance: appropriate follo...	2849

Measure ID	Measure Name	min
IMM_3	Healthcare workers given influenza vaccination	0
OP_18b	Average (median) time patients spent in the eme...	42
OP_18a	Average (median) time all patients spent in the...	44
OP_22	Left before being seen	0
SAFE_USE_OF_OPIOIDS	Safe Use of Opioids - Concurrent Prescribing	0
SEP_1	Appropriate care for severe sepsis and septic s...	0
SEV_SEP_3HR	Severe Sepsis 3-Hour Bundle	0
OP_18c	Average (median) time psychiatric/mental health...	41
SEV_SEP_6HR	Severe Sepsis 6-Hour Bundle	0
OP_29	Endoscopy/polyp surveillance: appropriate follo...	0

Measure ID	Measure Name	median
IMM_3	Healthcare workers given influenza vaccination	79.0
OP_18b	Average (median) time patients spent in the eme...	148.0
OP_18a	Average (median) time all patients spent in the...	154.0
OP_22	Left before being seen	1.0
SAFE_USE_OF_OPIOIDS	Safe Use of Opioids - Concurrent Prescribing	15.0
SEP_1	Appropriate care for severe sepsis and septic s...	65.0
SEV_SEP_3HR	Severe Sepsis 3-Hour Bundle	82.0
OP_18c	Average (median) time psychiatric/mental health...	247.0
SEV_SEP_6HR	Severe Sepsis 6-Hour Bundle	94.0
OP_29	Endoscopy/polyp surveillance: appropriate follo...	97.0

Measure ID	Measure Name	max
IMM_3	Healthcare workers given influenza vaccination	100
OP_18b	Average (median) time patients spent in the eme...	413
OP_18a	Average (median) time all patients spent in the...	422
OP_22	Left before being seen	23
SAFE_USE_OF_OPIOIDS	Safe Use of Opioids - Concurrent Prescribing	57
SEP_1	Appropriate care for severe sepsis and septic s...	100
SEV_SEP_3HR	Severe Sepsis 3-Hour Bundle	100
OP_18c	Average (median) time psychiatric/mental health...	6870
SEV_SEP_6HR	Severe Sepsis 6-Hour Bundle	100
OP_29	Endoscopy/polyp surveillance: appropriate follo...	100

Condition, measure, and state analysis

In [4]:
```
plt.figure(figsize=(12, 6))

df["Condition"].value_counts().head(12).sort_values().plot(
    kind="barh",
    color="steelblue"
)

plt.title("Number of Records by Healthcare Condition")
plt.xlabel("Number of Records")
plt.ylabel("Condition")
plt.tight_layout()
plt.show()

measure_counts = (
    df.groupby(["Measure ID", "Measure Name"])
      .size()
      .sort_values(ascending=False)
)

print(measure_counts.head(15))

state_counts = df["State"].value_counts().head(15)

state_counts.sort_values().plot(
    kind="barh",
    figsize=(10, 6),
    color="teal"
)

plt.title("States with the Most Hospital Measure Records")
plt.xlabel("Record Count")
plt.ylabel("State")
plt.tight_layout()
plt.show()

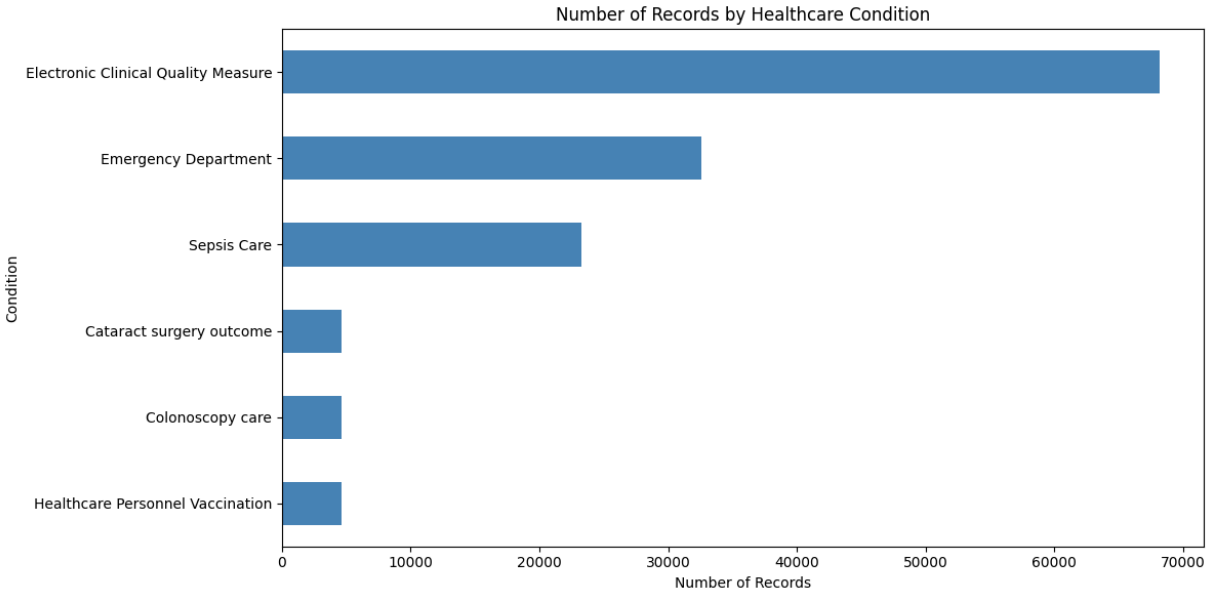
hospitals_by_state = (
    df.groupby("State")["Facility ID"]
      .nunique()
      .sort_values(ascending=False)
)
```

```
print(hospitals_by_state.head(15))

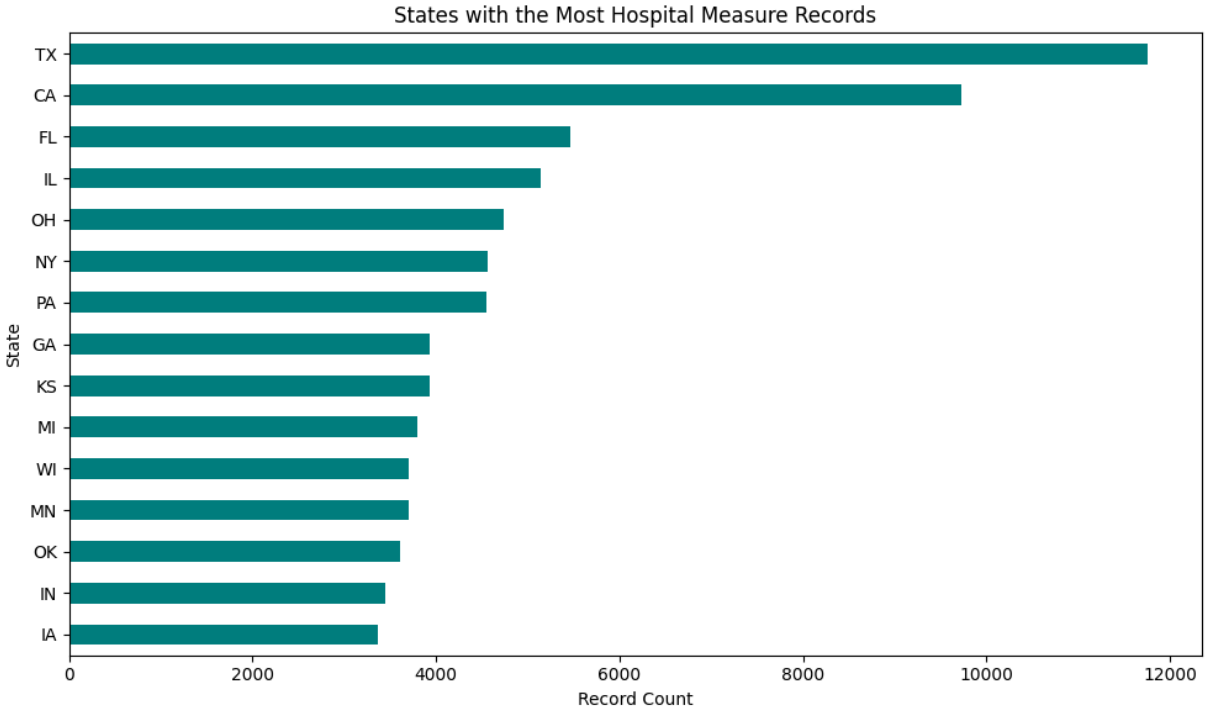
plt.figure(figsize=(10, 5))

sns.histplot(
    data=df,
    x="Score Numeric",
    bins=40,
    kde=True
)

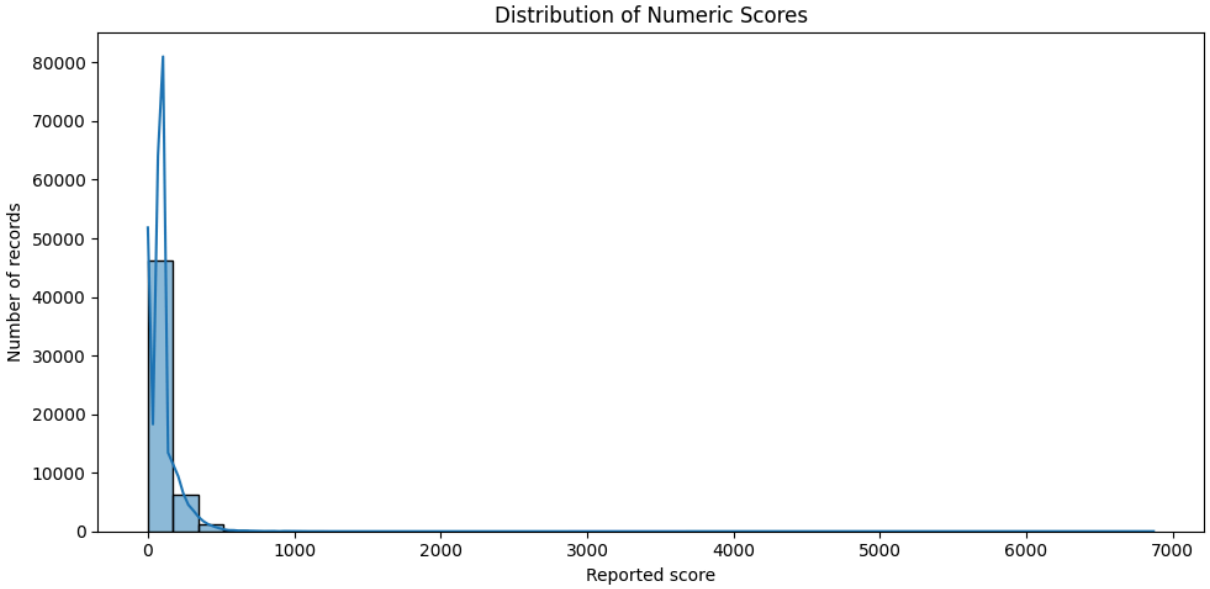
plt.title("Distribution of Numeric Scores")
plt.xlabel("Reported score")
plt.ylabel("Number of records")
plt.tight_layout()
plt.show()
```



```
Measure ID                                Measure Name
EDV                                       Emergency department volume
4658
GMCS                                      Global Malnutrition Composite Score
4658
GMCS_Malnutrition_Diagnosis_Documented  Global Malnutrition Composite Score: Malnutr
ition Diagnosis Documented
4658
GMCS_Malnutrition_Screening             Global Malnutrition Composite Score: Malnutr
ition Risk Screening
4658
GMCS_Nutrition_Assessment               Global Malnutrition Composite Score: Nutriti
on Assessment
4658
GMCS_Nutritional_Care_Plan              Global Malnutrition Composite Score: Nutriti
onal Care Plan
4658
HH_HYPER                                  Hospital Harm - Severe Hyperglycemia
4658
HH_HYPO                                   Hospital Harm - Severe Hypoglycemia
4658
HH_ORAE                                   Hospital Harm - Opioid Related Adverse Event
s
4658
IMM_3                                     Healthcare workers given influenza vaccinati
on
4658
OP_18a                                    Average (median) time all patients spent in
the emergency department before leaving from the visit, including psychiatric/mental
health patients and patients who were transferred to another facility. A lower numbe
r of minutes is better    4658
OP_18b                                    Average (median) time patients spent in the
emergency department before leaving from the visit, excluding patients transferred t
o another facility or psychiatric care/mental health patients. A lower number of min
utes is better             4658
OP_18c                                    Average (median) time psychiatric/mental hea
lth patients spent in the emergency department before leaving from the visit. A lowe
r number of minutes is better
4658
OP_18d                                    Average (median) time patients spent in the
emergency department before being transferred to another facility. A lower number of
minutes is better
4658
OP_22                                     Left before being seen
4658
dtype: int64
```




```
State
TX    401
CA    329
FL    184
IL    172
OH    160
NY    154
PA    153
GA    133
KS    132
MI    127
WI    124
MN    124
OK    122
IN    115
LA    112
Name: Facility ID, dtype: int64
```



OP_18a emergency department analysis

In [5]:
```
op18a = df[df["Measure ID"] == "OP_18a"].copy()

print("OP_18a records:", len(op18a))

print(
    op18a[
        ["Facility ID", "Facility Name", "State",
         "Measure Name", "Score", "Score Numeric",
         "Sample", "Footnote"]
    ].head(10)
)

op18a_numeric = op18a.dropna(
    subset=["Score Numeric"]
).copy()

print("Numeric OP_18a records:", len(op18a_numeric))

print(
    "Unique hospitals:",
    op18a_numeric["Facility ID"].nunique()
)

print(
    op18a_numeric["Score Numeric"].describe()
)

state_performance = (
    op18a_numeric.groupby("State")
    .agg(
        Hospital_Count=("Facility ID", "nunique"),
        Median_ED_Time=("Score Numeric", "median"),
        Mean_ED_Time=("Score Numeric", "mean")
    )
```

```
    .reset_index()
)

state_performance = state_performance.sort_values(
    "Median_ED_Time",
    ascending=True
)

print(state_performance.head(15))

plot_data = state_performance.nsmallest(
    15, "Median_ED_Time"
)

plt.figure(figsize=(10, 7))

sns.barplot(
    data=plot_data,
    x="Median_ED_Time",
    y="State",
    color="steelblue"
)

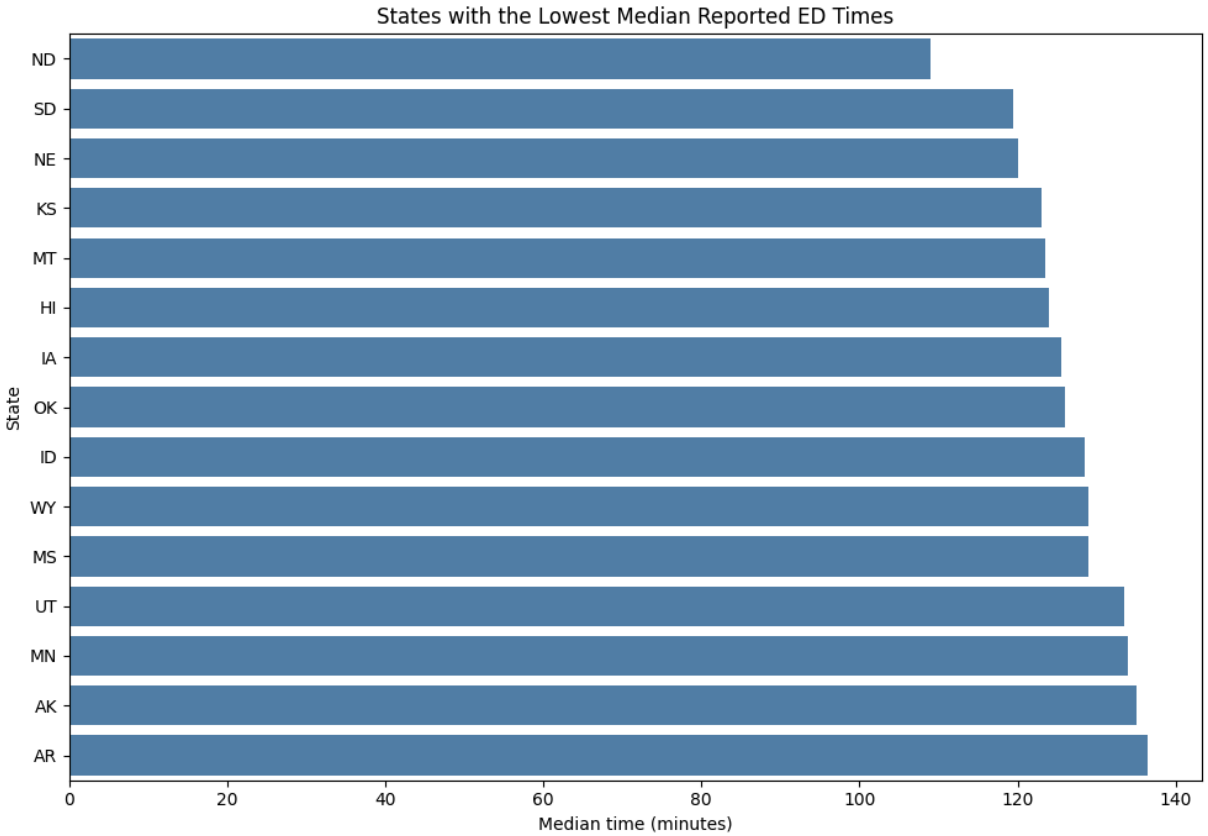
plt.title("States with the Lowest Median Reported ED Times")
plt.xlabel("Median time (minutes)")
plt.ylabel("State")
plt.tight_layout()
plt.show()
```

```
OP_18a records: 4658
    Facility ID                     Facility Name State  \
10       010001  SOUTHEAST HEALTH MEDICAL CENTER    AL
40       010005         MARSHALL MEDICAL CENTERS    AL
70       010006     NORTH ALABAMA MEDICAL CENTER    AL
100      010007         MIZELL MEMORIAL HOSPITAL    AL
130      010011               ST. VINCENT'S EAST    AL
160      010012   DEKALB REGIONAL MEDICAL CENTER    AL
190      010016    SHELBY BAPTIST MEDICAL CENTER    AL
220      010019             HELEN KELLER HOSPITAL    AL
250      010021               DALE MEDICAL CENTER    AL
280      010022    FLOYD CHEROKEE MEDICAL CENTER    AL

                                          Measure Name  Score  Score Numeric  \
10   Average (median) time all patients spent in th...    214            214
40   Average (median) time all patients spent in th...    146            146
70   Average (median) time all patients spent in th...    158            158
100  Average (median) time all patients spent in th...    137            137
130  Average (median) time all patients spent in th...    168            168
160  Average (median) time all patients spent in th...    174            174
190  Average (median) time all patients spent in th...    180            180
220  Average (median) time all patients spent in th...    189            189
250  Average (median) time all patients spent in th...    152            152
280  Average (median) time all patients spent in th...    127            127

     Sample Footnote
10      405     <NA>
40     1214     <NA>
70      376     <NA>
100    1103     <NA>
130     369     <NA>
160     410     <NA>
190     439     <NA>
220     373     <NA>
250     380     <NA>
280     436     <NA>
Numeric OP_18a records: 4052
Unique hospitals: 4052
count        4052.0
mean     162.846989
std       51.064954
min            44.0
25%           125.0
50%           154.0
75%          194.25
max           422.0
Name: Score Numeric, dtype: Float64
   State  Hospital_Count  Median_ED_Time  Mean_ED_Time
29    ND              43           109.0     120.27907
43    SD              50           119.5         118.8
30    NE              83           120.0    125.566265
17    KS             116           123.0    129.103448
27    MT              56           123.5    126.446429
12    HI              19           124.0         119.0
13    IA             112           125.5    132.660714
37    OK             101           126.0    128.475248
```

14	ID	38	128.5	135.078947
52	WY	23	129.0	133.304348
26	MS	67	129.0	137.313433
46	UT	44	133.5	140.409091
24	MN	120	134.0	143.55
0	AK	15	135.0	140.6
2	AR	66	136.5	143.045455



Outlier detection

In [6]:
```
q1 = op18a_numeric["Score Numeric"].quantile(0.25)
q3 = op18a_numeric["Score Numeric"].quantile(0.75)

iqr = q3 - q1

lower_bound = q1 - 1.5 * iqr
upper_bound = q3 + 1.5 * iqr

op18a_numeric["Outlier"] = (
    (op18a_numeric["Score Numeric"] < lower_bound) |
    (op18a_numeric["Score Numeric"] > upper_bound)
)

print("Lower bound:", lower_bound)
print("Upper bound:", upper_bound)
print("Flagged records:", op18a_numeric["Outlier"].sum())

outliers = op18a_numeric.loc[
    op18a_numeric["Outlier"],
```

```
    [
        "Facility ID", "Facility Name", "State",
        "Score Numeric", "Sample", "Footnote"
    ]
].sort_values("Score Numeric", ascending=False)

print(outliers.head(20))

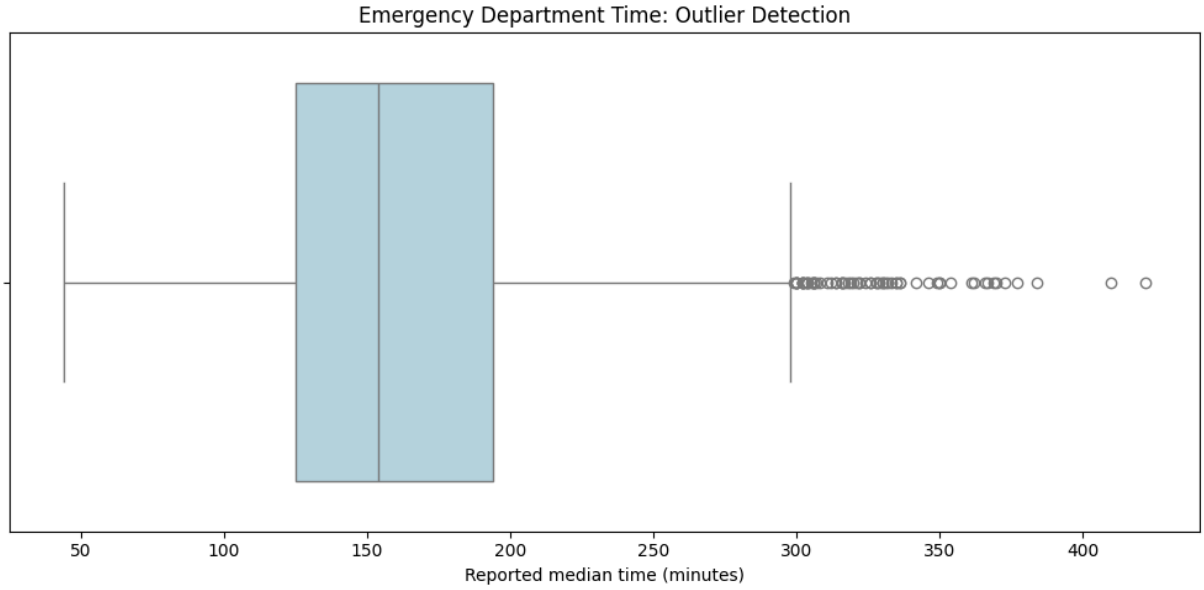
plt.figure(figsize=(10, 5))

sns.boxplot(
    data=op18a_numeric,
    x="Score Numeric",
    color="lightblue"
)

plt.title("Emergency Department Time: Outlier Detection")
plt.xlabel("Reported median time (minutes)")
plt.tight_layout()
plt.show()
```

```
Lower bound: 21.125
Upper bound: 298.125
Flagged records: 66
       Facility ID                                Facility Name State  \
106196      400114        MANATI MEDICAL CENTER DR OTERO LOPEZ    PR
12156       050327        LOMA LINDA UNIVERSITY MEDICAL CENTER    CA
115327      450213                    UNIVERSITY HEALTH SYSTEM    TX
21455       090011          MEDSTAR WASHINGTON HOSPITAL CENTER    DC
51492       190005       UNIVERSITY MEDICAL CENTER NEW ORLEANS    LA
115627      450289                                HARRIS HEALTH    TX
34267       140119              RUSH UNIVERSITY MEDICAL CENTER    IL
55962       210015      MEDSTAR FRANKLIN SQUARE MEDICAL CENTER    MD
56862       210062   MEDSTAR SOUTHERN MARYLAND HOSPITAL CENTER    MD
21365       090004      MEDSTAR GEORGETOWN UNIVERSITY HOSPITAL    DC
14565       050625                 CEDARS-SINAI MEDICAL CENTER    CA
35947       140281              NORTHWESTERN MEMORIAL HOSPITAL    IL
106550      400132       DOCTORS CENTER HOSPITAL  CAROLINA LLC    PR
21305       090001             GEORGE WASHINGTON UNIV HOSPITAL    DC
56262       210029        JOHNS HOPKINS BAYVIEW MEDICAL CENTER    MD
82270       330057                          ST PETER'S HOSPITAL    NY
59416       230046        UNIVERSITY OF MICHIGAN HEALTH SYSTEM    MI
58094       220086        BETH ISRAEL DEACONESS MEDICAL CENTER    MA
28237       110079                       GRADY MEMORIAL HOSPITAL    GA
70300       260105  SSM HEALTH SAINT LOUIS UNIVERSITY HOSPITAL    MO

       Score Numeric  Sample Footnote
106196            422    3341     <NA>
12156             410     376     <NA>
115327            384     382     <NA>
21455             377     375     <NA>
51492             373     435     <NA>
115627            370     368     <NA>
34267             369     395     <NA>
55962             367     357     <NA>
56862             366     364     <NA>
21365             362     400     <NA>
14565             361     402     <NA>
35947             354     400     <NA>
106550            350     407        3
21305             350     405     <NA>
56262             349     327     <NA>
82270             346     412     <NA>
59416             342     381     <NA>
58094             336     387     <NA>
28237             336     348     <NA>
70300             335     305     <NA>
```



Export cleaned data and analysis results

In [7]:
```
df.to_csv(
    "cms_hospital_quality_cleaned.csv",
    index=False
)

state_performance.to_csv(
    "ed_time_by_state.csv",
    index=False
)

outliers.to_csv(
    "ed_time_potential_outliers.csv",
    index=False
)
```

Key Findings

- Identified 4,658 records for the OP_18a emergency department measure, including 4,052 records with numeric scores.
- Calculated a median reported emergency department time of 154 minutes and a mean of approximately 162.8 minutes across numeric records.
- Found that North Dakota had the lowest state-level median reported time at 109 minutes, followed by South Dakota at 119.5 minutes.
- Flagged 66 potential outlier records using the IQR method for further investigation.

Outcome

- Cleaned hospital quality dataset

- State-level emergency department performance analysis
- Potential outlier records
- Click here to open the Dashboard Created on DataBricks
- Tableau dashboard: View Dashboard